Manufacturing Process Executive Summary

Key Performance Indicators

Simulation LOTs: 5,000

Normal LOTs: 4,321 (86.42%)

Defect LOTs: 679 (13.58%)

Overall Average Yield: 85.11%

Normal LOT Average Yield: 98.48%

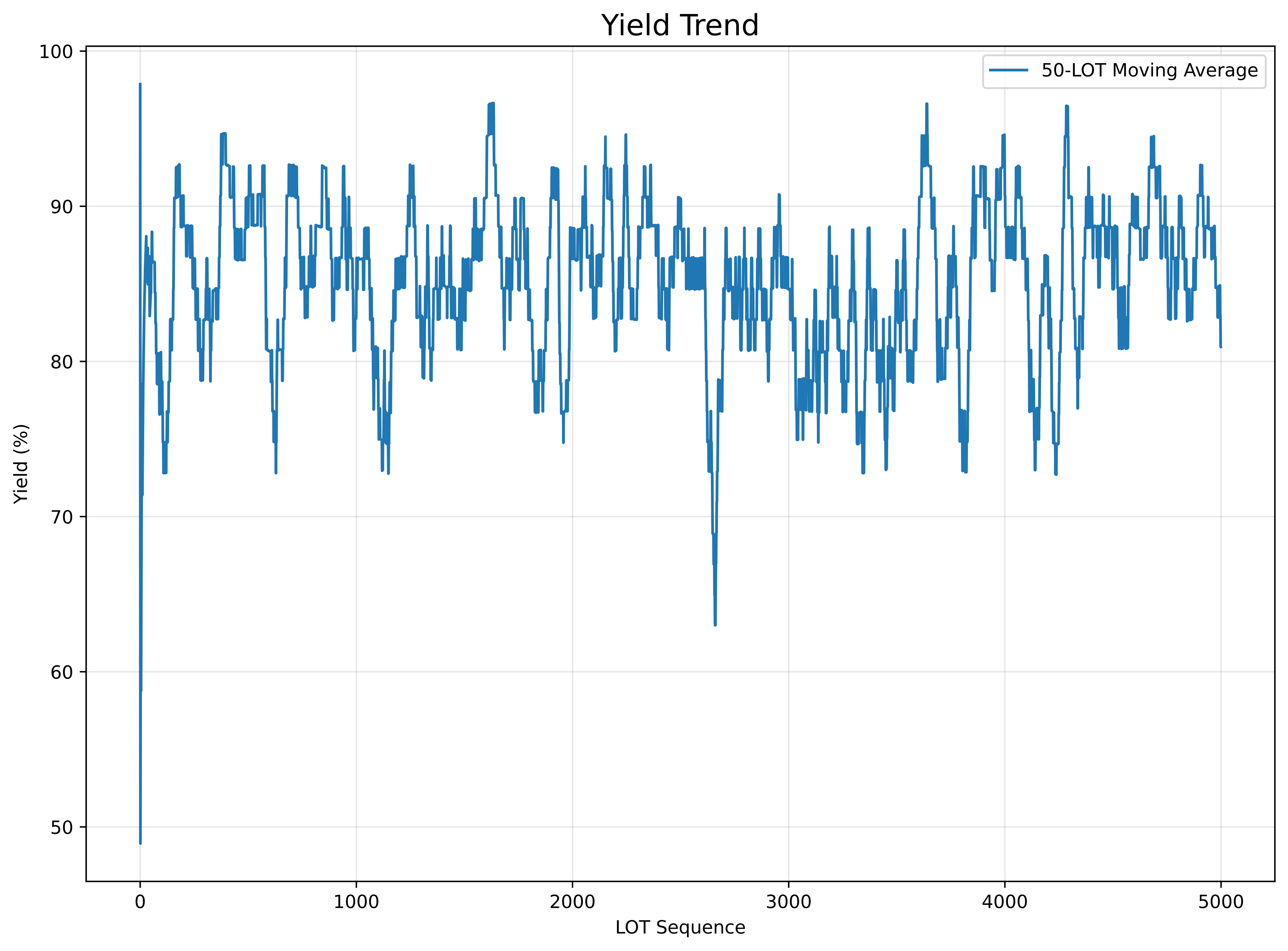
Defect Breakdown

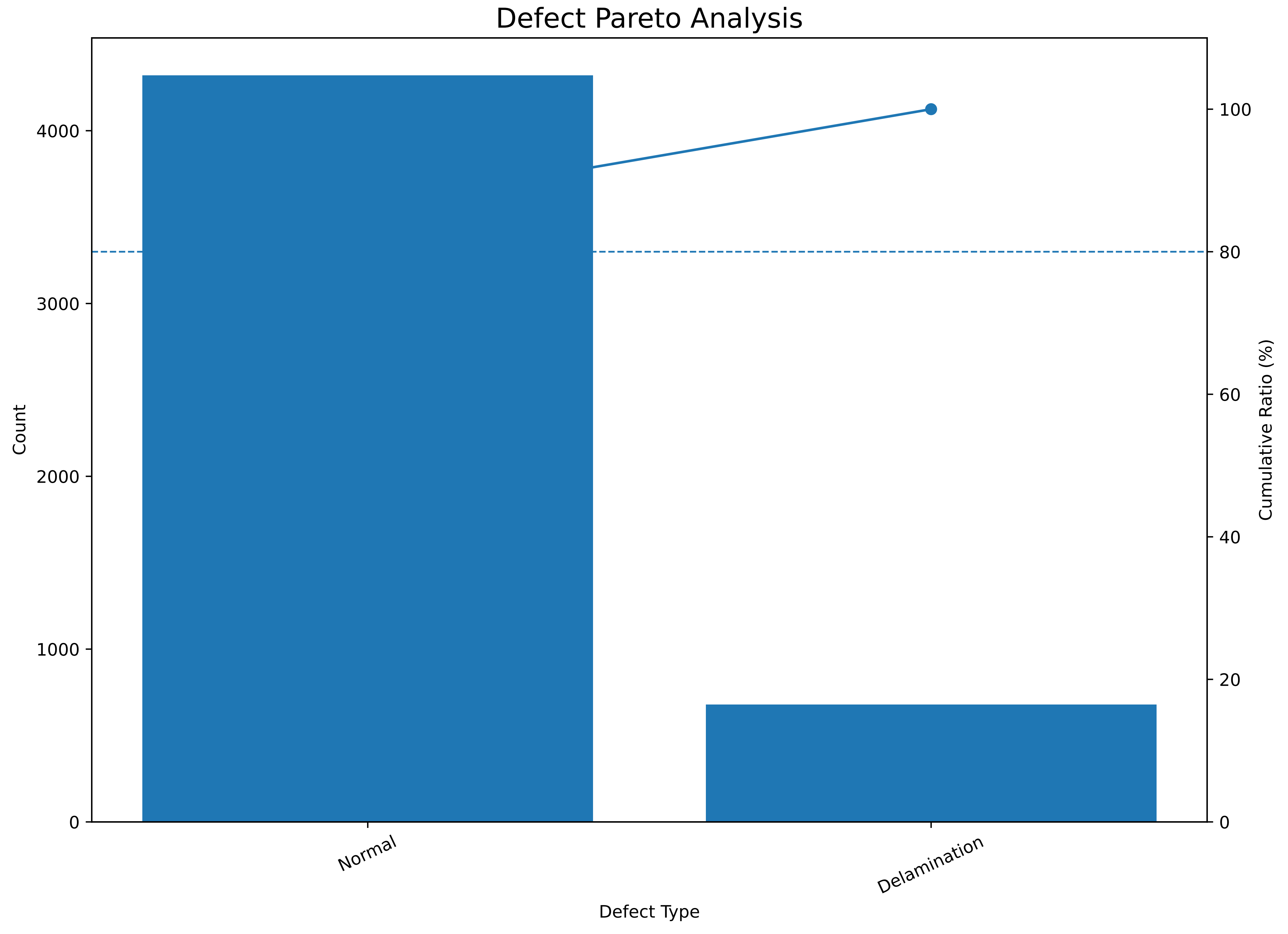
- Delamination: 679 LOTs (13.58%)

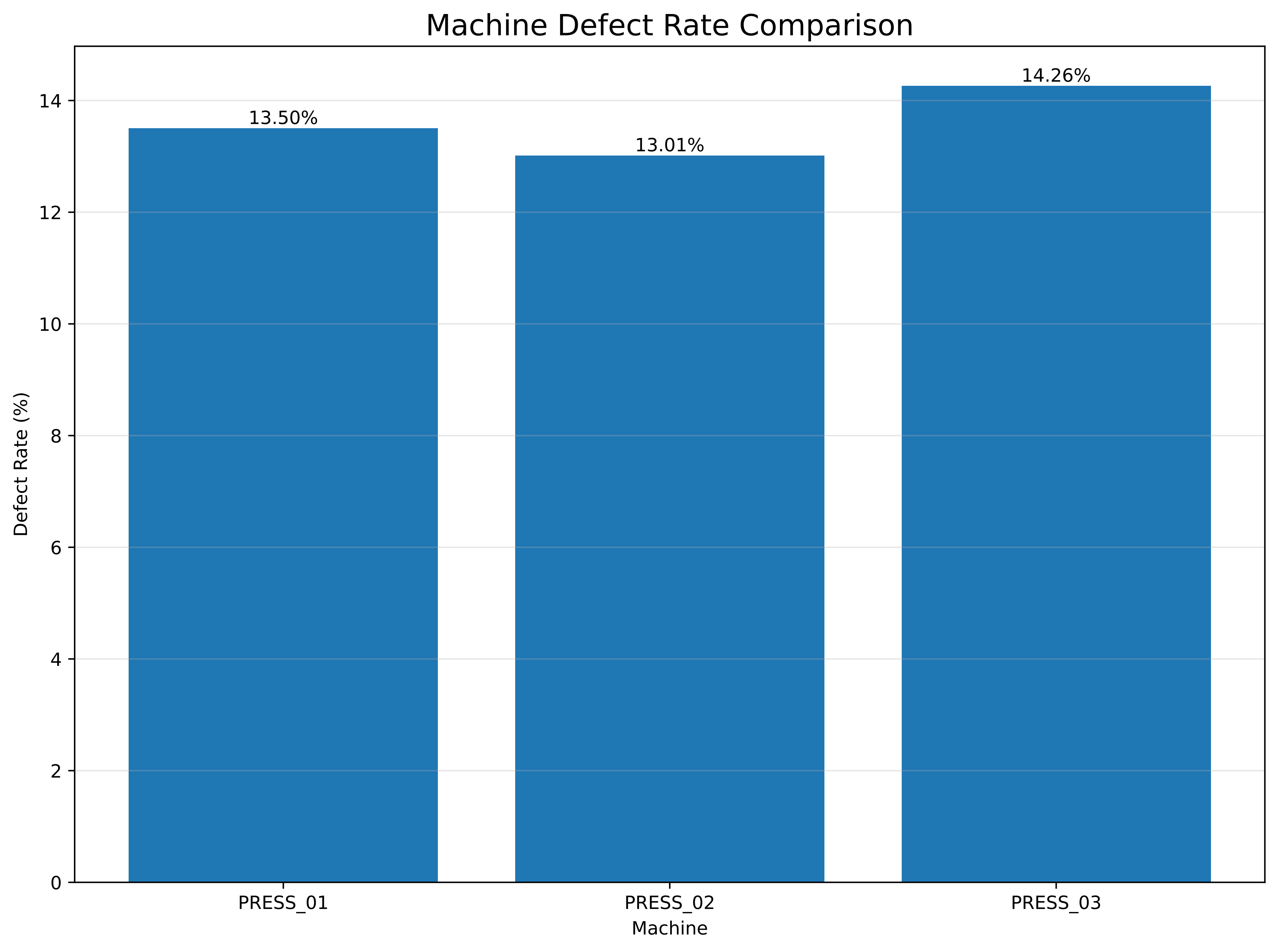
Conclusion

Based on the current 5,000-LOT simulation, the expected overall yield is 85.11% and the expected defect rate is 13.58%. The largest defect category is Delamination, 679 LOTs.

Note: The values above are simulation results generated from the current process model, not actual production forecasts.


Yield Trend
50-LOT Moving Average
Yield (%)
100
90
80
70
60
50
0
1000
2000
3000
4000
5000
LOT Sequence


Defect Pareto Analysis
Count
Cumulative Ratio (%)
Defect Type
Normal
Delamination
0
1000
2000
3000
4000
0
20
40
60
80
100


Machine Defect Rate Comparison
13.50%
13.01%
14.26%
Defect Rate (%)
0
2
4
6
8
10
12
14
PRESS_01
PRESS_02
PRESS_03
Machine

Process Capability Summary

Parameter	Mean	Std	LSL	USL	Index	Value	Assessment
CZ_Roughness	349.941	6.775	320.0	380.0	Cpk	1.473	Capable
Total_Thickness	32.001	0.368	30.0	34.0	Cpk	1.811	Excellent
Peel_Strength	499.781	20.004	350.0	nan	Cpl	2.496	Excellent
ABF_Roughness	349.817	6.701	300.0	400.0	Cpk	2.478	Excellent

Assessment guide: Excellent >= 1.67, Capable >= 1.33, Marginal >= 1.00, Improvement Required < 1.00

Cpk is used for two-sided specifications. Cpl or Cpu is used for one-sided specifications.

Process Adjustment Recommendations

Recommended Adjustment Priorities

1. CZ_Concentration: Increase toward target 10.000 %
2. Anneal_Temp: Reduce toward target 200.000 °C
3. Anneal_Time: Reduce toward target 90.000 min
4. Cure_Temp: Reduce toward target 150.000 °C
5. Cure_Time: Reduce toward target 60.000 min

Priority	Parameter	Current_Mean	Normal_Mean	Defect_Mean	Target	Direction
1	CZ_Concentration	10.0	10.004	9.972	10.0	Increase
2	Anneal_Temp	199.981	199.961	200.112	200.0	Decrease
3	Anneal_Time	90.055	89.925	90.88	90.0	Decrease
4	Cure_Temp	150.001	149.989	150.079	150.0	Decrease
5	Cure_Time	60.001	59.982	60.125	60.0	Decrease

Method: Recommendations are ranked using the standardized difference between normal and defect LOTs combined with Random Forest feature importance.

Important: These are directional recommendations derived from synthetic simulation data. They must not be applied directly to production without DOE, engineering review, and process validation.

Quality Parameter Summary

Parameter	count	mean	std	min	25%	50%	75%	max
CZ_Roughness	5000.0	349.941	6.775	321.32	345.29	349.885	354.512	375.94
Total_Thickness	5000.0	32.001	0.368	30.57	31.752	32.004	32.248	33.281
Peel_Strength	5000.0	499.781	20.004	421.68	486.328	499.73	513.32	572.46
ABF_Roughness	5000.0	349.817	6.701	326.59	345.29	349.66	354.26	375.01
Yield	5000.0	85.105	33.746	0.0	97.72	98.34	98.85	100.0